

Recycling carbon tax for inclusive green growth: A CGE analysis of India

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ABSTRACT

In this decade, India has been pursuing a low carbon inclusive growth strategy. However, carbon tax, the most direct price instrument to reduce carbon emissions, has not found favour with policymakers because of its supposed detrimental effects on economic growth and income distribution. In the Indian context, the literature indicates that though carbon tax is extremely effective in abating carbon emissions, it simultaneously leads to reductions in GDP. There is, thus, an undesirable trade-off between economic growth and climate change mitigation. However, in trying to overcome this trade-off through a double-dividend from carbon tax, these studies have not really explored all possible options. Whether the carbon tax will yield a double-dividend or not, will depend upon how the carbon tax revenue is recycled.

The present paper fills this gap in the literature on recycling carbon tax for inclusive green growth by exploring the consequences of using carbon tax revenue for investment to build capacity in all sectors or exclusively in the clean energy sectors and to execute transfers to households to improve the distribution of income. This analysis has been done with a recursively dynamic India-specific CGE model having a disaggregated energy sectors and an endogenous income distribution module.

1. Introduction

A carbon tax is a tax on the carbon content of fuels (mainly, coal, oil and natural gas) that emit carbon dioxide emissions when burnt. Carbon tax is the most direct price instrument to reduce carbon emissions. However, such a carbon tax has not found favour with policymakers in India because of its supposed detrimental effects on economic growth and income distribution. Even the literature on carbon tax in India has not been encouraging for the policymakers (Pal et al., 2015; Ojha, 2009a,b; Fisher-Vanden et al., 1997). The principal finding that emerges from these studies is that even though carbon tax is extremely effective in abating carbon emissions, it simultaneously leads to reductions in GDP. There is, thus, an undesirable trade-off between economic growth and climate change mitigation. However, in trying to overcome this trade-off through a double-dividend from carbon tax, these studies have not really explored all possible options. Whether the carbon tax will yield a double-dividend or not, will depend upon how the carbon tax

revenue is recycled. It is usual to consider two types of recycling of carbon tax, known as the revenue neutral and the revenue positive types (Pal et al., 2015). In the former type, while carbon tax is introduced, some other direct/indirect tax is removed or reduced to maintain revenue neutrality; in the latter type, the (additional) carbon tax revenue reaped by the government from the introduction of carbon tax is mostly recycled to the household sector to augment its personal income. In both cases, there are reductions in carbon emissions but the impact on GDP is an adverse one, as shown by Pal et al. (2015), Ojha (2009a,b) and Fisher-Vanden et al. (1997). The impact on income distribution is also mildly unfavourable (Pal et al. (2015); Ojha, 2009a,b).¹ Ojha (2009a,b) considers another variant of the revenue positive type recycling wherein the additional carbon tax revenue is transferred to the households with a larger proportion accruing to the poorer households. Predictably, there are some poverty alleviating effects of this mode of recycling, but overall GDP losses resulting from the carbon tax fail to get reversed. In short, double-dividend, in the typical sense of achieving GDP gains and carbon

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¹ The CGE model in the paper by Fisher-Vanden et al. (1997) is based on only a single representative household, and, therefore, cannot capture the impact of a carbon tax policy on income distribution. On the other hand, the CGE model in Ojha (2009a,b) has multiple rural and urban consumption expenditure household groups and that in Pal et al. (2015) has multiple rural and urban socioeconomic household groups so as to reflect the distributional implications of a carbon tax policy.

emissions abatement simultaneously, does not seem to come about in these India-specific studies. To our best knowledge, none of these and other carbon tax studies for India such as, McKibbin (2004), Bussolo & O'Connor (2001), Jayadevappa and Chhatre (1996) and Shah and Larsen (1992) have found any possibility for a double dividend. This, however, has not deterred policymakers in emerging economies from continuing their search for not only a double dividend, but, in fact a triple dividend in terms of GDP gains (growth), reduced income inequality (inclusive) and carbon emissions abatement (green) from a carbon tax.²

The present study fills this gap in the literature on recycling carbon tax for inclusive green growth by exploring the consequences of using carbon tax revenue for investment to build capacity in all sectors or exclusively in the clean energy sectors and to execute transfers to households to improve the distribution of income. This analysis has been done with a recursively dynamic India-specific CGE model having a highly disaggregated energy sector. Moreover, the CGE model incorporates an endogenous income distribution module designed to capture the distributional impacts of the suggested carbon tax policies. In a dynamic setting, while the carbon tax *per se* abates emissions, recycling the additional carbon tax revenue into investment may well foster growth and reduce inequality of income. At least, the possibility of a triple dividend (or a double dividend) from a carbon tax needs to be explored before ruling it out.

In the next section, we briefly discuss the Indian experience with carbon taxes. Section 3 is dedicated to an exposition of the CGE model employed in the present study. Sections 4 and 5 portray the baseline scenario, and the carbon tax policy scenarios respectively. The sensitivity analysis of our model is also presented in section 5. Section 6 concludes with policy implications and suggestions for future research.

2. India's experience with carbon taxes for tackling climate change

Since the beginning of this decade, India has been pursuing a low carbon inclusive growth (LCIG) strategy (Government of India, 2011). This strategy is a multi-pronged one. The main objectives in this comprehensive strategy are (i) enhancing energy efficiency, (ii) promotion of use of clean coal, (iii) augmenting production of other greener non-fossil-fuel based energy, such as, solar, wind, nuclear and hydropower, (iv) creating carbon sinks, (v) encouraging fuel-switching towards relatively cleaner sources of energy. The types of policy instruments employed for achieving these objectives are broadly four: (i) regulations and standards, such as, building codes, bio-fuel standards and vehicle-efficiency standards; (ii) public funding for relevant R&D; (iii) awareness and capacity building activities; and (iv) market-based instruments, which, in turn, include quantity instruments (emission trading schemes and renewable energy certificates) and price instruments (removal of subsidies and imposition of carbon taxes). Two important schemes based on quantity instruments have been launched in India to create carbon markets. These are (i) Perform, Achieve and Trade (PAT) scheme, and (ii) Renewable Energy Certificate (REC) scheme. While PAT scheme aims to improve energy efficiency in industries by trading in energy efficiency certificates in energy-intensive sectors, the REC scheme, as a major initiative under the National Solar Mission (NSM) of the National Action Plan for Climate Change (NAPCC), Government of India (2008) seeks to address the mismatch between the availability of renewable energy sources and the requirement of obligated entities to meet their renewable purchase obligations. However, it is the price instrument, carbon tax, which remains the most potent market-based policy tool for inducing fuel-switching towards cleaner

sources of energy.

Although explicitly India does not have any carbon tax till date, implicitly or *de facto* it has a carbon tax in the form of a coal cess since 2010, which in fact has been rising in magnitude.³ This clean environment cess, it may be noted, is over and above the 5 per cent – 6 per cent excise duty on coal which predates the launch of climate change mitigation policies. In July 2017, after the unveiling of the goods and services tax (GST) reforms, this excise duty on coal has been replaced with a GST at the rate of 5 per cent. There is no clean environment cess on other fossil fuels – petrol and diesel, but excise duties on these two fuels have always been there, not as a climate change mitigating policy, but as a major revenue generator for the government. These excise duties continue even after GST reforms as fuel is out of the purview of the system, and their rates are continuously rising. Even though originally not intended to be so, excise duties on petrol or diesel, also act as implicit carbon taxes, by effectively putting a price in emissions. These taxes generate a price signal to reduce the fuel burnt, and thereby the CO₂ emissions. Concomitantly, there would be effects on GDP. The reason for preferring an implicit tax on carbon is not explicitly stated in the policy documents of Government of India (GoI). It is not likely that the implicit carbon tax through cesses and excise duties on fossil fuels will produce better results in terms of reduction in carbon emissions and changes in GDP than an explicit carbon tax. Indeed, the reverse is likely to be true. The greater effectiveness of explicit carbon tax *vis-à-vis* implicit carbon taxes has been shown to hold by Yokoyama et al. (2000) and Lachapelle and Sustainable Prosperity (2011). Moreover, pricing carbon through an explicit tax on carbon content of fuels rather than through cess and/or excise duties on fuels is likely to be more “economically efficient”, as the latter will unavoidably bear the imprint of policymakers’ (distortionary) discretion. It would seem, therefore, that the real reason for the adoption of the implicit carbon tax in India is a pragmatic one. A tax on the carbon content of the fuels is a novelty and therefore a difficulty that the tax-collecting machinery would best postpone if not avoid. The implicit carbon tax therefore becomes a default option within the existing mode of functioning of the fiscal machinery of India. However, this is likely to change in an economy continually moving on a path of economic reforms (Government of India, 2015; Shakti Sustainable Energy Foundation et al., 2018).

Turning to options in recycling revenues generated through carbon pricing, it may be noted that theoretically the range here is rather wide. Among the wide-ranging recycling options for carbon tax revenues, the important ones, in our view, are (i) distributing them uniformly to all household groups across-the-board or preferentially to the low income/consumption expenditure household groups to compensate for the burden borne by them due to the imposition of the carbon tax in the first place, (ii) investing additionally in various sectors of the economy for capacity expansion, (iii) investing exclusively in clean (renewable) energy sectors, and (iv) investing in research and development for enhancing energy efficiency. In this paper, we focus on the first three, leaving out the last one for a separate study. Needless to say, these options are not mutually exclusive. They may all be exercised simultaneously with varying degrees of emphases, as reflected in the share of the carbon tax revenue dedicated for each option. In practice, the range of options are even wider because they are driven by contingencies faced by the policymakers. This is more so in case of India (Narain, 2015; Bhuyan, 2016). However, it is precisely in striking a balance between exigency and vision (shaped by theory) that the challenge lies in policy making.

In India, the coal cess was introduced in 2010, and, around the same time, the national clean energy fund (NCEF) was created and financed primarily by the revenues from the former. The purpose of NCEF was to fund research for improved energy efficiency and innovative clean

² Hereinafter, double dividend would mean garnering any two of the following three expected benefits from carbon tax: (i) GDP gains, (ii) better income distribution, (iii) reduced carbon emissions abatement.

³ Government of India (2015); Government of India, Union Budget, various years.

Table 1

Sectors and Household classes of our Model.

Sl. No.	Sectors of our Model	Sl. No.	Sectors of our Model
1	Agriculture	15	Wind power
2	Jatropha (feedstock for biodiesel)	16	Solar power
3	Forestry	17	Fertilizer
4	Biomass	18	Chemicals
5	Coal	19	Other energy intensive sector
6	Crude oil	20	Machinery
7	Gas	21	Other Intermediates
8	Petroleum products	22	Construction
9	Biodiesel	23	Consumer goods
10	Thermal power	24	Land transport
11	Electricity: Petroleum-based	25	Rail transport
12	Electricity: Gas-based	26	Other Transport
13	Hydropower	27	Health,
14	Nuclear power	28	Education
		29	Services
Household Socio-economic Classes			
Sl. No.	Rural	Sl. No.	Urban
1	Rural non-agricultural self-employed (RH1)	6	Urban self-employed (UH1)
2	Rural agricultural labour (RH2)	7	Urban salaried class (UH2)
3	Rural other labour (RH3)	8	Urban casual labour (UH3)
4	Rural agricultural self-employed (RH4)	9	Urban other households (UH4)
5	Rural other households (RH5)		

energy projects of public sector or private sector, using the revenues from this cess. The NCEF, nourished by the cess, has been very useful in funding clean energy projects in early years of the last decade (Panda and Jena, 2012). Of late with worsening of fiscal situation of the central government, there has been a tendency to use this fund for non-energy and non-environmental purposes.⁴

If coal cess is a *de facto* carbon tax, then quantitatively speaking, coal cess should be translatable into an equivalent carbon tax. Government of India (2014) does precisely that and converts the specific coal cess into equivalent carbon tax (using standard carbon emission factors and net calorific values of coal). In 2015, the GoI revised its coal cess from Rs. 50 per ton to Rs. 100 per ton. Translating this into carbon tax equivalent shows that a carbon tax of US \$ 0.50 in 2014 was doubled to US\$ 1.00 in 2015, and presently it is Rs. 400 per tonne.

With these high cesses on coal, it is understandable that India now claims to have implicit carbon tax rates which are higher than international carbon tax rates, and, hence, asserts that its commitment to reduce carbon emissions is much greater than is acknowledged globally. While there is an element of truth in this assertion, the fact is that carbon tax policy in India has evolved (and is still evolving) haphazardly toward a carbon taxation regime which lacks overall consistency and does not seem to be linked to specific carbon emission targets. In other words, Indian policy makers having resisted a direct carbon tax for its likely adverse impact on GDP, have eventually admitted an implicit carbon tax through the backdoor. But this type of carbon taxation limited to coal, penalises the carbon emitted from coal but spares that from other fossil fuels – e.g., refined oil and natural gas. Such an indirect carbon tax cannot induce efficient fuel-switching away from fossil fuels towards cleaner sources of energy. In our view, the transition to a direct carbon tax linked to declared carbon emission targets is a natural next step in the evolution of carbon tax policy in India which should not be delayed anymore.

⁴ For instance, the fund been used to compensate states for revenue loss post-GST (see The Hindu, 2019).

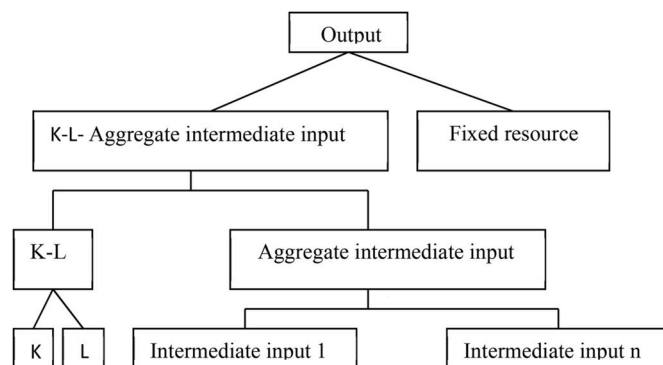


Fig. 1. Production structure of fossil fuel sectors (Coal, Crude Oil and Natural Gas).

3. CGE model structure

The model used in this study is a single country, multi-sectoral, neoclassical-type, price-driven, recursively dynamic CGE model with a disaggregated energy sector. As Table 1 indicates, the model consists of 29 sectors, with disaggregated electricity sector (7 in all). There are two factors of production (capital and labour), three types of domestic institutions (households, enterprises and government), and an external sector capturing the transaction with the rest of the world. Furthermore, there are nine types of households or socio-economic groups in our model economy of India based on their location (rural/urban) and type of occupation (see Table 1).⁵

The production structure of the fossil fuel sectors (Coal, Oil and Gas) takes into account the limited availability of these resources. In case of the fossil fuel sectors (Fig. 1), the top nest is a Leontief aggregation of the capital-labour-aggregate intermediate input composite and the fixed fossil fuel resource. The capital-labour-aggregate intermediate composite is a Leontief function of the capital-labour composite and aggregate intermediate inputs bundle, which in turn is a Leontief aggregate of all individual commodities serving as intermediate inputs. The capital-labour composite (aggregate value added) is in turn a constant elasticity of substitution (CES) aggregation of capital and labour. A CES function implies substitution between two inputs, at a constant rate, due to changes in relative prices of the inputs.

In case of the non-fossil fuel sectors (Fig. 2), the top nest is a Leontief function of the energy-capital-labour composite and the non-energy intermediate inputs aggregate. The energy-capital-labour composite in turn is a CES function of the energy composite and the capital-labour composite. The energy composite is a CES function of the non-electric composite and the electric composite. The non-electric composite is a CES aggregation of coal, oil, gas, refined oil and biodiesel. The electric composite is a CES aggregation of renewable electricity composite and non-renewable electricity composite. The renewable electricity composite is a CES aggregation of hydro, nuclear, wind and solar, while the non-renewable electricity composite is a CES aggregation of coal, oil and gas based electricity.⁶ The capital-labour composite is a CES function of capital and labour. The non-energy intermediate input bundle is a

⁵ In our single-country model, the entire country (India) as a whole is represented.

⁶ The electricity sector nesting structure is based on that of the EMPAX-CGE model (2008), with the elasticities at the various levels of the electricity sector nest chosen in such a manner as to reproduce the sectoral outputs of 2007-08 Social accounting Matrix (SAM) of the Indian economy. That is to say, many trial runs were done with initial substitution elasticities taken from either EMPAX-CGE (2008) or OECD-Env-Linkages-CGE (2014), and, finally, the chosen substitution elasticities were the ones which helped replicate the sectoral outputs of the India SAM.

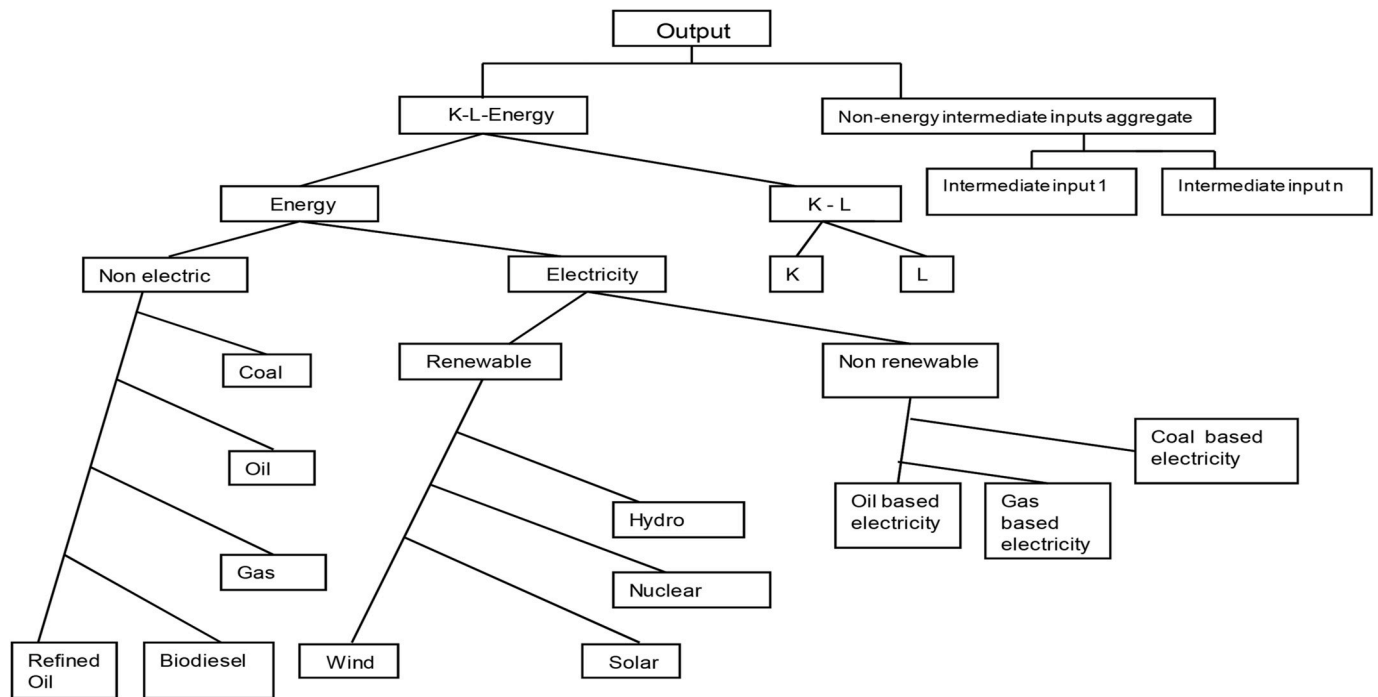


Fig. 2. Production structure of non-fossil-fuel sectors.

Leontief function of all the non-energy intermediate inputs. The nested CES production structure allows substitution between energy, capital, and labour on one hand, and between renewable and non-renewable sources of electricity on the other, in response to changes in relative prices. Finally, in accordance with the basic tenets of neoclassical economics, producers are assumed to maximize profits or minimise costs while operating in perfectly competitive markets.

This disaggregation of households captures differences between the different socio-economic groups in terms of income (and sources of income) and consumption patterns. Households maximize utility subject to income and prices, and the household demand for commodities is modelled through the linear expenditure system (LES).⁷ Household income comprises of income derived from labour and capital, and transfers from the government and rest of the world. Households also save part of their income and pay taxes to the government.

Government expenditure is on the consumption of goods and services, transfers to households and enterprises, and subsidies. Government income is from taxes (direct and indirect), capital, public and private enterprises, and rest of the world. In our model, direct tax implies income tax on household and corporate tax on enterprises. The indirect taxes include excise duty, import and export tariffs, sales, stamp, service, and other indirect taxes. Government savings which is the difference between government expenditure and income is determined residually in the model.

Imperfect substitution between domestically produced and imported goods is allowed for in CGE models. In other words, producers/consumers are free to sell or buy goods from the domestic or foreign market based on relative prices. The Armington function is used to capture the substitution possibilities between domestically produced and imported goods. The import demand function, derived from the Armington function, specifies the value of imports based on the ratio of domestic

and import prices. The constant elasticity of transformation (CET) function, on the other hand, is used to capture substitution possibilities between domestic and foreign sales. The export supply function, derived from the CET function, specifies the value of exports based on the ratio of domestic and export prices. Both the Armington and CET functions are of the CES type. The world prices are assumed to be fixed in the model.

The model is Walrasian in character, that is markets for all commodities and factors clear through adjustment in prices. The consumer price index (CPI) is chosen as the *numeraire*, and is therefore fixed. In other words, all income and prices in the model are relative to the CPI. The macro closures play an important role in CGE models. The closures are specified taking into account the macro economic reality of the country or region being modelled. Our model follows an investment-driven closure, keeping in view the prime role played by public investment in sponsoring growth in a mixed economy like India. The saving-investment balance is maintained through adjustments in aggregate savings (sum of household, government, enterprise and foreign savings). Both foreign savings and the real exchange rate are endogenous. The government consumption expenditure is fixed within a period, and government savings is residually determined. Both direct and indirect tax rates are fixed. The household savings rate is fixed while the household consumption expenditure and income are endogenously determined in the system. Finally, we assume in equilibrium, demand for both the factors of production (labour and capital) matches respective supplies. While labour is allowed to move across sectors, sectoral capital is fixed, within a period (year).

3.1. Income distribution

Factor incomes (factor prices times the respective factor demands) are readily generated in a CGE model, but factor income distribution is not likely to reflect accurately the personal or household income distribution, because most household income groups in real-world economies derive their incomes from both labour and capital. Most CGE models concerned with income distribution, therefore, map factor incomes onto household incomes. The degree of detail in the classification of households into groups or classes is typically governed by the kind of

⁷ In LES demand function, one assumes that consumer has a subsistence demand of each commodity depending on its income, which needs to be met first. After that being met, the rest of the consumption expenditure is allocated across various commodities based on choice function and demand elasticity of the commodities.

data that is available for the country in question. As noted earlier, there are nine socioeconomic classes in our model. These nine groups, arranged in the order of increasing per-capita incomes are: RH2 (rural agricultural labour), RH4 (rural agricultural self-employed), UH3 (urban casual labour), RH1 (rural non-agricultural self-employed), RH3 (other rural labour), UH1 (urban self-employed labour), UH2 (urban salaried labour), RH5 (other rural households) and UH4 (other urban households). According to our constructed SAM, RH2, RH4, UH3 and UH2 depend mostly on labour for their incomes, RH1, RH3 and UH1 earn their incomes more or less evenly from capital and labour, and finally, RH5 and UH4 derive their incomes mainly from capital.

3.2. Carbon emissions

CO₂ is emitted due to the burning of fossil fuels (coal, gas and crude oil) and due to very process of output generation.⁸ Finally, CO₂ emissions also result from the final consumption of households and the government. We use fixed CO₂ emission coefficients to calculate the sector-specific CO₂ emissions from each of the three sources of carbon emissions.⁹ For the total CO₂ emissions generated in the economy, we first aggregate the emissions from each of the sources over the 29 sectors and subsequently sum up the aggregate emissions across the three sources.

3.3. Carbon taxes

Carbon taxes are applied on the consumption of fossil fuels in the economy. Carbon taxes are based on the proportion of each fuel's CO₂ emissions, that is, rupees per ton of CO₂ emitted by the fuel. The carbon tax rate multiplied by a sector's CO₂ emission gives the emission tax payments by that sector. Summing across sectors we get the total carbon tax payments, which is then recycled to the household sector as additional transfer payments by the government. In the BAU scenario, the carbon tax rate is fixed at zero and there are, therefore, no carbon tax payments. It may be noted that, the producer's cost function is modified to include the emission taxes so that these taxes induce a substitution in favour of lower carbon-emitting fossil fuels. A carbon tax is translated into price increase for each of the fossil fuels – coal, crude oil and natural gas. The price increase is maximum for coal which has the highest CO₂ emission factor, followed by refined oil and natural gas. In response, a cost minimising (or a profit maximising) producer changes the fossil fuel input mix away from coal and towards refined oil and natural gas, and also from fossil fuel inputs towards non-fossil-fuel inputs.

3.4. Dynamics

The model is multi-period in nature, where the unit of period is one year. It is a recursively dynamic model and is solved as a sequence of static single-year CGE models, after updating sectoral capital stocks and available labour supply each year. The sectoral capital stocks are exogenously given at the beginning of a particular period. Between two periods, there will be additions to capital stocks in each sector because of the investment undertaken in that sector in the previous period. More precisely, the sectoral capital stocks for any year t are arrived at by adding the investments by sectors of destination, net of depreciation, in year $t-1$ to the sectoral capital stocks at the beginning of the year $t-1$. Likewise, labour supply is updated each year by adding the new entrants to the labour force.

The complete set of equations of our CGE model are delineated in Ojha et al. (2020). Computationally, the model has been solved in General Equilibrium Modelling Systems (GAMS).

⁸ For example, the cement sector (a part of the *other energy intensive industries* in our sectoral classification) releases CO₂ in the limestone calcination process.

⁹ The emission coefficients are drawn from Ojha et al. (2009a).

3.5. Assumptions regarding technological change

The baseline (BAU) growth trajectory of the economy is created by assuming exogenously specified growth in total factor productivity (TFP). Based on the TFP growth estimates of Pal et al. (2015), the annual TFP growth rates for the 29 sectors in our CGE model are calibrated and they vary in the range of 1–3 per cent.

An autonomous energy efficiency growth rate is assumed to capture improvements in energy efficiency in the future. As Ojha, Pohit & Pal (2009b) indicate, India's track record in energy efficiency improvement has been very good. Hence, most studies project autonomous energy efficiency improvement (AEEI) growth rate in India to be 1.5% or above per annum. However, we have assumed the annual AEEI growth rate to be 1.5% per annum in this paper following the study by Ojha, Pohit & Pal (2009a).

3.6. Database

Our CGE model has been calibrated to the benchmark equilibrium data set of the Indian economy for the (financial) year 2007–2008 obtained from the Social Accounting Matrix (SAM) for that year, which we prepared ourselves for this study following the methodology outlined in Pal et al. (2012). The key structure of Indian economy in the base year 2007–08 is shown in Table A2-1 (see Ojha et al., 2020, pp. 7). As this table shows, agriculture and services are two most important contributors of value-added in the base year.

Most of the parameters used in the model are derived from SAM constructed for this table. The values of the important parameters are shown in Ojha et al. (2020). For example, Table A2-2 in Ojha et al. (2020, pp. 8) shows the sector-wise applied tax/subsidy rate used in our model. The savings rate, income tax rate, shares of government transferred used in our model economy are shown in Table A2-3. On the other hand, Armington elasticity parameters for our model are shown in Table A2-4. These are drawn from Ojha, Pohit, Pal (2009a). Table A2-5 displays values of the LES parameters used in our model (see Ojha et al., 2020, pp. 10).

In our model, we have used fixed CO₂, N₂O and CH₄ emission coefficients to calculate sector-specific GHG emissions from each of the three sources of carbon emissions.¹⁰

An important driver of our model is growth rate of labour force. The time series data on population for the time period 2007–08 to 2040–41 is available from CENSUS of India. To estimate the labour supply for India we have used the data on Labour force Participation Rate (LFPR) of India, obtained from NSSO report (Government of India). The growth rate comes to about 1.9 per cent per annum. The other important exogenous variable of our model is quantum of annual net inflow of foreign savings. To obtain time-series data on foreign savings we used the projected growth rate from the macro-econometric model of the Indian economy prepared by NCAER (2006). With regard to government consumption expenditure, we assume that it remains stable at 0.07% of GDP based on past 5 year average preceding the base year 2007–08.

Given the benchmark data set for all the variables and the elasticity parameters, the *shift* and *share* parameters are calibrated in such a manner that if we solve the model using the base-year data inputs, the

¹⁰ The emission coefficients are drawn from (INCCA (2010)).

Table 2
Stylised facts of BAU scenario.

Variables	Year								
	2007	2010	2012	2015	2020	2025	2030	2035	2040
GDPFC (Rs. billion)	44372	54464	62654	74321	98510	130301	171999	226208	294508
Growth Rate	6.83	6.83	7.00	5.69	5.64	5.59	5.55	5.48	5.28
CO ₂ (million tonnes)	1322	1468	1593	1733	2104	2454	3007	3681	4025
CO ₂ /GDP (tonnes/1000Rs.)	0.03	0.03	0.03	0.02	0.02	0.02	0.02	0.02	0.01
Energy use/GDP (KWH/Rs.)	0.13	0.10	0.10	0.09	0.09	0.08	0.07	0.06	0.05
GINI Coefficient	0.37	0.40	0.41	0.42	0.42	0.42	0.43	0.43	0.42
Trade balance/GDP ratio	−0.06	−0.05	−0.05	−0.05	−0.06	−0.05	−0.03	0.06	0.07
Household consumption/GDP ratio	0.58	0.53	0.53	0.54	0.56	0.57	0.57	0.47	0.49

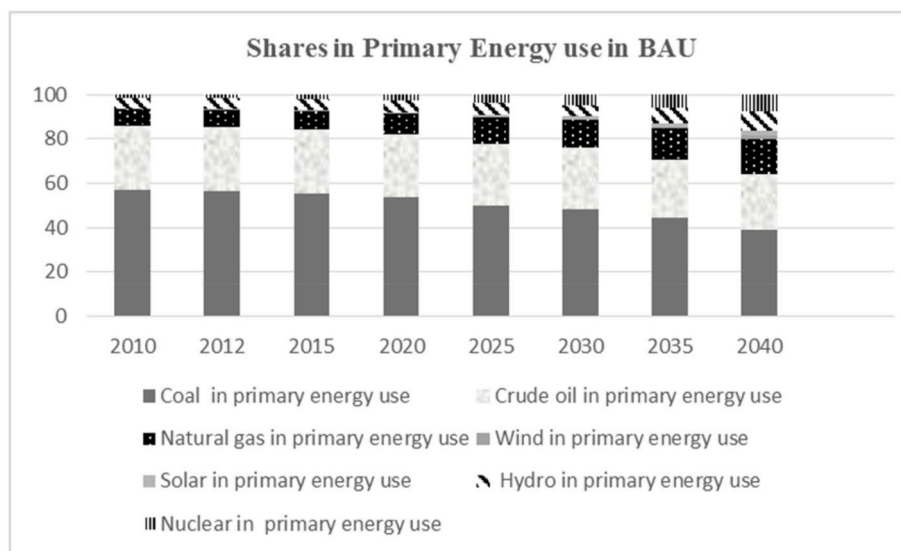


Fig. 3. Shares in Primary Energy use in BAU.

result will be the input data itself.¹¹ Basically it means that we replicate the data in the SAM through our model.

Finally, using a time series of the exogenous variables of the model, we generate a sequence of equilibria for the period 2007 to 2040 (i.e., 2007–08 to 2040–41). From the sequence of equilibria, the growth paths of selected (macro) variables of the economy are outlined to describe the base-line scenario, spanning the 34-year time interval, 2007 to 2040.

4. Business-as-Usual (BAU) scenario

In the baseline scenario GDP grows at an average of 5.74 percent per annum between 2007 and 2040 while CO₂ emissions grow at an average of 3.37 percent per annum during the same period (Table 2). The energy intensity of GDP decreases by more than 60 percent during this period while the CO₂ intensity of GDP decreases by more than half (Table 2). The reduction in CO₂ intensity of GDP is mainly driven by the replacement of CO₂ intensive coal by relatively cleaner sources of energy. There is significant increase in the shares of natural gas, hydro, nuclear and renewables (wind and solar) in total primary energy use between 2007

and 2040 (Table 2). It is interesting to note that the CO₂ intensity of GDP decreases by about 30 per cent between 2007 and 2020 in the BAU scenario, which is consistent with India's policy objective (Copenhagen Pledge) of a 20–25 per cent reduction in CO₂ intensity of GDP between 2005 and 2020. The implication is that India is in a position to meet its stated policy objective with respect to CO₂ intensity reduction without much effort. The energy mix in the BAU scenario is shown in Fig. 3, which is similar to the energy mix scenario developed by Shukla et al. (2008).

5. Policy scenarios

The climate change mitigation agenda in India has multifarious elements comprising of command and control, market-based and hybrid types of policy instruments, not all of which can be taken up for analysis in a single paper. Our focus in this paper is mainly on the impact of carbon tax variously recycled on carbon emissions abatement, GDP, and income distribution. We have undertaken the following five policy simulations with varying modes of recycling carbon tax revenue as indicated below:

(1) Policy scenario 1 (HH transfer):

In this scenario, the revenue from carbon tax (tax on CO₂ emissions) is recycled to different household groups based on the shares of government transfers these households receive in the BAU scenario.

(2) Policy scenario 2 (Inv-All):

¹¹ The shift and share parameters used in the production functions (and Armington functions) of a CGE model are calibrated so that the CGE model solution replicates the initial equilibrium database of the SAM (see pp. 42, Burfisher, 2016). As an example, let us consider a CGE model (like ours) that uses CES production functions: $Q = A [\alpha K^{-\theta} + (1-\alpha)L^{-\theta}]^{-1/\theta}$ where 'Q' is output, 'K' is capital input, 'L' is labor input; 'θ' is the substitution parameter, 'A' is the efficiency parameter, and 'α' is the 'distribution' parameter; 'A' and 'α' are respectively referred to as the 'shift' and 'share' parameters in the CGE parlance.

Table 3

Per cent changes in Key variables in Policy Scenario 1: HH transfer relative to BAU.

Year	GDP	Total Emissions	CO2 Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.01	-2.97	0.00	-1.39	-0.16	0.67	0.18
2030	-0.05	-5.12	-5.00	-1.59	-0.22	1.42	0.23
2035	-0.13	-5.73	-5.26	-1.79	-0.42	-1.58	0.46
2040	-0.20	-8.80	-11.76	-1.96	-0.81	-2.87	0.87
Average	-0.06	-3.77	-3.67	-1.12	-0.27	-0.39	0.29

The carbon tax revenue is recycled towards investment in the different sectors with the allocation proportions being the same as that in the BAU scenario.

(3) Policy scenario 3 (Inv-All & HH transfer):

The scenario 3 is a mix of scenario 1 and scenario 2. That is, 50 per cent of the carbon tax revenue is allocated towards investment following the rule of scenario 2, while the remaining 50 percent is transferred to household groups with the allocation principle being the same as in scenario 1.

(4) Policy scenario 4 (Inv-CE):

In scenario 4, the carbon tax revenue generated is recycled towards investment in the clean energy sectors (wind, hydro, solar, and nuclear) with sectoral allocation proportions being 25 per cent each.

(5) Policy scenario 5 (Inv-CE & HH transfer):

In scenario 5, the 60 per cent of the carbon tax revenue generated is recycled towards investment in the clean energy sectors (wind, hydro, solar, and nuclear) with each, receiving 15 per cent of the carbon tax

revenue, and the remaining 40 per cent being transferred to the different household groups based on the shares of government transfers these households receive in the BAU scenario.

The carbon tax rates in all the above scenarios 1 were obtained by running the model in reverse mode, that is, by making the carbon tax rates endogenous for specified reductions in CO₂ emissions relative to the baseline. The carbon tax rates are for a 5 per cent reduction in CO₂ emissions relative to the baseline for the period 2021-30, and for a 10 per cent reduction in CO₂ emissions relative to the baseline for the period 2030-40. Carbon tax was introduced in the year 2021 because it was observed in the BAU scenario that India can achieve its policy objective of 20–25 per cent reduction in CO₂ intensity by 2020, simply with trend rate of improvement in energy intensity and carbon intensity of GDP without requiring any carbon tax till 2020. Thereafter, the average carbon tax rates obtained by running the model in the reverse mode are: Rs. 66 per tonne of CO₂ between 2021 and 2025, Rs. 98 per tonne of CO₂ between 2026 and 2030, Rs. 216 per tonne of CO₂ between 2031 and 2035, and Rs. 502 per tonne of CO₂ between 2036 and 2040. Further, the revenue from carbon tax (tax on CO₂ emissions) is recycled to different household groups based on the shares of government transfers these households receive in the BAU scenario.

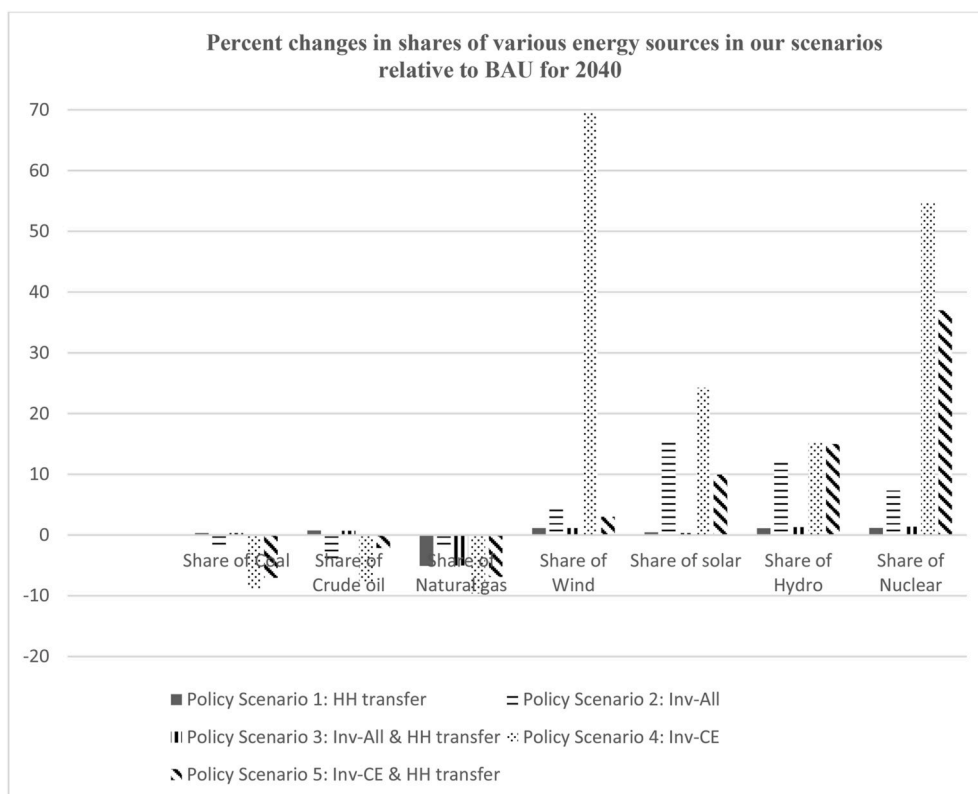


Fig. 4. Percent changes in shares of various energy sources in our scenarios relative to BAU for 2040.

Table 4

Per cent changes in Key variables in Policy Scenario 2: Inv-All relative to BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	-1.16	0.00	0.00	0.00
2025	0.02	-2.89	-5.26	-1.33	0.02	0.89	0.01
2030	0.01	-4.99	-5.88	-1.54	0.03	1.54	0.02
2035	0.03	-5.60	-6.25	-1.82	0.04	-1.78	0.03
2040	0.09	-8.52	-7.14	-3.92	0.06	-2.44	0.05
Average	0.02	-3.67	-4.09	-1.63	0.02	-0.29	0.02

Table 5

Per cent Changes in Key variables in Policy Scenario 3: Inv-All & HH transfer relative to BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2025	-0.01	-2.89	0.00	-1.39	-0.06	0.78	0.09
2030	-0.03	-5.02	-5.00	-1.59	-0.07	1.48	0.12
2035	-0.04	-5.65	-5.26	-1.79	-0.15	-1.68	0.25
2040	-0.04	-8.65	-11.76	-1.96	-0.30	-2.65	0.46
Average	-0.02	-3.70	-3.67	-1.12	-0.10	-0.35	0.15

5.1. Results and discussion: policy scenario 1 (HH transfer)

A carbon tax results in price increases for each of the fossil fuels – coal, refined oil and natural gas. The extent of price increase in case of each of these fuels is determined by the carbon content of the respective fuels. The price increase is largest for coal because coal has the highest carbon content, and smallest for natural gas which has the lowest carbon content, and none for the renewables. Producers respond by switching from coal towards refined oil, natural gas and renewables as a source of energy. At the same time, higher (average) energy prices force a reduction in overall energy use. Carbon emissions are reduced on account of both fuel switching and overall reduction in fuel use. Since the substitution elasticities across the various sources of energy are low, the fuel reducing effect dominates over the fuel switching effect, resulting in GDP losses. This is broadly the mechanics at work in most CGE models which have shown GDP loss arising out of a carbon tax. The recycling of carbon tax revenue to households will boost consumption demand in the economy to some extent, but that will not have the familiar Keynesian expansionary effects on output because ours is full-employment model. Instead, there will changes in relative prices across goods and factors of production – some relative prices will rise while some others will fall. (Increases in output can come through creation of additional capacity by augmenting investment in the economy, which will happen, as we shall see, in the next policy simulation.)

In this scenario both the energy intensity and the CO₂ intensity of GDP are lower relative to the BAU scenario (Table 3). While the energy intensity of GDP is lower by about 1.12 per cent on an average, CO₂ intensity of GDP is lower on an average by 3.67 per cent. The total emissions also decline on an average by 3.77 per cent. The decline in CO₂ intensity of GDP occurs because of the decline in the energy intensity of GDP (the fuel reducing effect) and also because of fuel switching. The share of fossil fuels, such as, natural gas in primary energy use decrease, while the shares of cleaner energy sources like hydro, nuclear, wind and solar increase (Fig. 4).¹²

While the total emissions decline by 3.77 per cent, GDP declines by only 0.06 per cent relative to the BAU scenario. The Gini coefficient of

income inequality relative to the BAU scenario is lower on an average by 0.27 per cent. Trade deficit-GDP ratio worsens mildly by 0.64 per cent on an average because carbon taxes induce some switching from domestically produced commodities to imported substitutes, hence, imports increase. Household consumption-GDP ratio increases despite the decline in GDP, because there are transfer payments funded by the carbon tax revenues.

In brief, this scenario shows that by recycling carbon tax to the various household groups, it is possible to bring down total emissions, energy intensity of GDP, carbon intensity of GDP and income inequality significantly, while bearing a very minor loss in GDP.

5.2. Results and discussion: policy scenario 2 (Inv-All)

Carbon taxes in single-period CGE models tend to reduce GDP because, as argued above, the fuel-reducing effect dominates the fuel-switching effect. However, in a sequentially dynamic CGE model like ours, the accretions in capital stocks resulting from the investments made in the previous year – apart from that in labour stocks on account of the increase in population – are a key driver of economic growth. Under such circumstances, ploughing back carbon tax revenue into investment will reverse the initial GDP losses into gains after sufficient time has elapsed. Moreover, the smaller the carbon tax, the sooner will be this turnaround. In our model, the carbon tax rate of Rs. 66 per tonne introduced in year 2021 being very small in magnitude, the turnaround comes about in the year 2023.

In this scenario, GDP for the 20-year period (2021–2040) increases marginally by 0.02 percent because the ploughing back of carbon tax revenue into investment augments the productive capacity (capital stocks) of the economy (Table 4). Total carbon emissions decrease on an average by 3.67 per cent. However, there is a minor increase in income inequality relative to BAU on an average by 0.02 percent. Trade deficit-GDP ratio also increases but it is too small an increase to compromise macroeconomic stability. CO₂ intensity of GDP decreases relative to BAU on an average by 4.09 percent, while the energy intensity of GDP declines relative to BAU on an average by 1.63 percent. The shares of Coal, Crude Oil and Natural Gas decline in total primary energy use, while the shares of Hydro, Nuclear, Wind and Solar show an increase (Fig. 3). The main message from this scenario is that if the revenue garnered from carbon tax is recycled into investment then there is a double benefit from the former – i.e., there are gains in the form of lower CO₂ emissions and higher GDP, but in income distribution, there is a small deterioration.

¹² The share of fossil fuels, such as, coal and crude oil, increase but only marginally. The reason is: (1) coal is also used as a medium of cooking for some household classes whose income rises (note fall in GINI coefficient). Since crude oil is the source of transport fuel in India, the demand for same may rise in spite of fall in CO₂ intensity of GDP.

Table 6

Per cent Changes in Key variables in Policy Scenario 4: INV-CE) relative to BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.01	-3.26	-5.26	1.33	0.01	-0.42	-0.01
2030	0.00	-5.75	-5.88	3.08	0.01	-1.00	-0.01
2035	0.00	-6.74	-6.25	9.09	0.01	0.77	-0.01
2040	0.00	-10.51	-14.29	15.69	0.02	1.04	-0.01
Average	0.00	-4.38	-5.28	4.86	0.01	0.07	-0.01

5.3. Results and discussion: policy scenario 3 (Inv-All & HH-transfer)

Scenario 3 is a combination of scenarios 1 and 2. That is, 50 per cent of the carbon tax revenue is allocated towards investment following the rule of scenario 2 while the remaining 50 percent is transferred to household groups, the allocation principle being the same as in scenario 1.

There is an average decline in GDP of 0.02 per cent in this scenario, whereas there was an increase in GDP of 0.02 per cent in Scenario 2 and a decrease in it of 0.06 per cent in scenario 1 (see Table 5). Total emissions decline on an average by 3.70 per cent. Cash transfers to households out of the carbon tax revenues helps household consumption-GDP ratio to go up by 0.15 per cent. The increase of 0.61 per cent in trade deficit-GDP ratio is a non-issue from the macroeconomic stability viewpoint. With the Gini coefficient in this scenario being lower than in BAU on an average by 0.10 per cent, income inequality in Scenario 3 is showing signs of improvement. It may be recalled that income inequality in Scenario 2 was found to worsen. On the other hand, income inequality in policy scenario 1 showed distinct improvement. The outcome for income inequality in this scenario is a resultant of the two opposing forces on income distribution active in the scenarios 1 and 2.

CO₂ intensity of GDP on average declines by 3.67 per cent in this scenario, while the corresponding figure is 4.09 per cent in Scenario 2 (Tables 4 and 5). The energy intensity of GDP declines in Scenario 3 by 1.12 per cent on an average. The share of fossil fuels, such as, natural gas in primary energy use decrease, while the shares of cleaner energy sources like hydro, nuclear, wind and solar increase (Fig. 4).

In this scenario, there is a small reduction in GDP. However, on every other count – total CO₂ emissions, CO₂ emissions intensity, energy intensity, Gini coefficient – there is a clear gain.

5.4. Results and discussion: policy scenario 4 (Inv-CE)

Scenario 4 is similar to scenario 2 except that carbon tax revenue is now invested equally towards the clean energy sectors (wind, solar, hydro, nuclear), rather than all sectors of the economy. In other words, in this scenario we study the effects of augmenting investments in the clean energy sectors of the economy, and financing them through the revenues earned from the levy of a carbon tax.

In Scenario 4, a scrutiny of GDP levels of scenario 4 and the BAU scenario reveals minuscule increases in GDP in the former. However, in percentage terms (up to second decimal place), there is no noticeable change in GDP relative to BAU (Table 6). Even in the Gini coefficient of

income inequality, there are very small increases in this scenario as compared to the baseline scenario.

Total emissions decline on average by 4.38 per cent. CO₂ intensity of GDP on average also declines in Scenario 4 by 5.28 per cent while the corresponding figure is 4.09 per cent in Scenario 2. In this scenario the energy intensity of GDP rises by 4.86 per cent on average, while there is a decline of 1.63 percent in Scenario 2 (see Tables 4 and 6). The shares of coal, crude oil, and natural gas expectedly decline in primary energy use while the shares of wind, solar, hydro and nuclear show an increase (Fig. 3).

It follows that, investment in clean energy sectors is beneficial in terms of climate change mitigation (i.e., reduction in carbon emissions), but that is all. Energy is not used frugally, but profligately because energy prices decline due to the boost in investment in the clean energy sectors. Decline in carbon emissions comes about from shifting the sourcing of energy towards renewables rather than fossil fuels. Income distribution worsens.

5.5. Results and discussion: policy scenario 5 (Inv-CE & HH transfer)

In this scenario, while total carbon emissions decline by 4.13 per cent, GDP declines marginally by 0.02 per cent (Table 7). However, household consumption-GDP ratio increases because of the transfer payments. There is marginal decline in income inequality in this scenario. The implication is that recycling of revenues to households is beneficial from an equity perspective but not from an efficiency perspective. The average decline in the CO₂ intensity of GDP of 5.28 per cent is similar to that of Scenario 4 (Table 6), while the average increase in energy intensity of GDP of 2.80 per cent is lower in this scenario relative to scenario 4 where it is 4.86 per cent (Table 6). The shares of coal, crude oil, and natural gas in primary energy use decline while the shares of wind, solar, hydro and nuclear show an increase (Fig. 3). In sum, this scenario fares well in terms of reducing carbon emissions as well as carbon intensity of GDP. Income distribution also shows improvement. But, energy intensity rises, and there are GDP losses.

5.6. Impact on income distribution across policy scenarios

As we have seen above, the income inequality across nine household groups (captured by the Gini coefficient, ignoring within-group inequality) worsens in scenarios 2 and 4, but diminishes in scenarios 1, 3 and 5. Worsening income distribution in scenarios 2 and 4 is linked to the fact of additional investment made in these scenarios. Increased investment leads to a rise in the price of capital goods, because the

Table 7

Per cent Changes in Key variables in Policy Scenario 5: Inv-CE & HH transfer relative to BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.00	-3.14	-5.26	0.00	-0.01	0.02	0.07
2030	-0.02	-5.52	-5.88	1.54	-0.01	-0.03	0.09
2035	-0.04	-6.36	-6.25	5.45	-0.03	-0.17	0.18
2040	-0.05	-9.76	-14.29	9.80	-0.07	-0.62	0.34
Average	-0.02	-4.13	-5.28	2.80	-0.02	-0.13	0.11

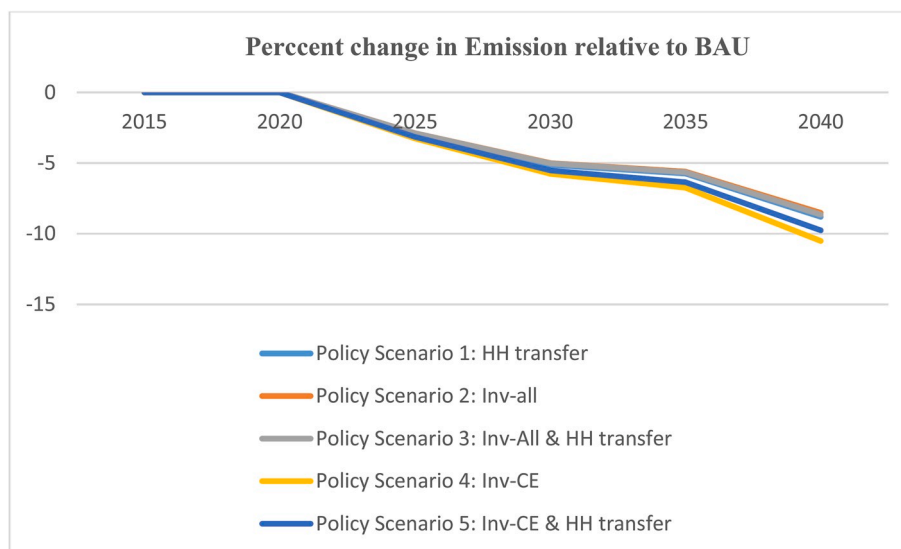


Fig. 5. Percent change in Emission relative to BAU.

Table 8

Changes in Key variables relative to BAU across Scenarios (average percent change).

Key Variables	Policy Scenario 1: HH transfer	Policy Scenario 2: Inv-All	Policy Scenario 3: Inv-All & HH transfer	Policy Scenario 4: Inv-CE	Policy Scenario 5: Inv-CE & HH transfer
GDP	-0.06	0.02	-0.02	0.00	-0.02
CO2 Intensity of GDP	-3.67	-4.09	-3.67	-5.28	-5.28
Energy Intensity of GDP	-1.12	-1.63	-1.12	4.86	2.80
Gini Coefficient	-0.27	0.02	-0.10	0.01	-0.02
Share of Coal	0.32	-1.83	0.33	-8.81	-7.03
Share of Crude oil	0.75	-3.82	0.70	-7.88	-2.14
Share of Natural gas	-5.09	-2.02	-5.02	-9.61	-6.90
Share of Wind	1.15	4.28	1.14	69.67	3.00
Share of solar	0.45	15.39	0.34	24.24	9.93
Share of Hydro	1.13	11.98	1.29	15.11	14.99
Share of Nuclear	1.16	7.30	1.36	54.80	36.97
Trade balance/GDP	-0.39	-0.30	-0.35	0.07	-0.13
Household consumption/GDP	0.29	0.02	0.15	-0.01	0.11

Note: Authors' estimate.

Lower value of Gini Coefficient implies lower income inequality.

demand-generating effect of the former is stronger than capacity-creating effect (which takes longer to materialise). The rise in the return on capital relative to labour benefits the groups RH5 and UH4 most, because they derive their incomes mainly from capital and the groups, RH1, RH3 and UH1, somewhat less, as they have even dependence on capital and labour for their incomes, but adversely affects the groups, RH2, RH4, UH3 and UH2, which depend primarily on labour. These losing groups (except for UH2) also occupy the lowest rungs in the income ladder. Not surprisingly, income inequality is enhanced in scenarios 2 and 4. On the other hand, transfer payments are being made in scenarios 1, 3 and 5 to the household groups which create demand mostly for labour intensive goods. So, there is a rise in the relative price of labour *vis-à-vis* capital and that benefits the low-income groups (RH2, RH4, UH3) more than the high-income groups (RH5 and UH4). Hence, income inequality reduces.

5.7. Overview of results across the five policy scenarios

A comparison of the effects on different economic and energy related indicators in the five policy scenarios reveals the commonalities and differences among them. All the five policy scenarios bring down total carbon emissions though by different amounts. The reduction in emissions across the five policy scenarios relative to BAU can be seen in

Fig. 5. They also differ in their changes in emission and energy intensities and in their impacts on GDP. Using the effect on GDP, as the key differentiator, we compare the policy scenarios below (Table 8).

Scenarios 1, 3 and 5 are characterized by marginally negative effects on GDP relative to BAU. However, all of these three policy scenarios show an improvement in income distribution. These three policy scenarios also show a lower carbon emission intensity in comparison to that of BAU. However, energy intensity relative to BAU is lower for scenarios 1 and 3, but higher for scenario 5. In terms of total emissions and emission intensity, policy option 5 scores better than policy options 1 and 3, but it is clearly an inferior option *vis-a-vis* options 1 and 3 in terms of energy intensity. A policy option which uses more rather than less energy per unit of GDP is not likely to find favour with policy makers. Hence, the choice really is between policy options 1 and 3. In terms of reductions in energy intensity and CO₂ intensity, these two policy options are equivalent. In terms of total emissions abatement and improvement in income distribution, policy option 1 scores slightly better than policy option 3, but, with respect to loss in GDP, former is inferior to the latter. Hence, policy makers in emerging economics (who instinctively resist reductions in GDP) are likely to prefer policy option 3 over policy option 1.

Scenarios 2 and 4, show positive GDP gains. However, these gains are significant only in the former; in the latter, GDP gains are miniscule.

Table 9

Per cent changes in Key variables in S1-Policy Scenario:1 relative to S1-BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2025	0.01	-2.96	0.00	-1.39	-0.16	0.66	0.18
2030	-0.05	-5.11	-5.00	-1.57	-0.22	1.40	0.24
2035	-0.14	-5.71	-5.25	-1.78	-0.42	-1.56	0.48
2040	-0.21	-8.78	-11.77	-1.93	-0.81	-2.86	0.88
Average	-0.06	-3.77	-3.67	-1.11	-0.27	-0.39	0.29

Table 10

Per cent changes Key variables in in S1-Policy Scenario:2 relative to S1-BAU.

Year	GDP	Total Emissions	CO ₂ Intensity	Energy Intensity	Gini Coefficient	Trade balance/GDP ratio	HH consumption/GDP ratio
2015	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2020	0.00	0.00	0.00	-1.15	0.00	0.00	0.00
2025	0.02	-2.89	-5.28	-1.31	0.02	0.88	0.02
2030	0.01	-4.96	-5.91	-1.54	0.03	1.52	0.02
2035	0.02	-5.57	-6.28	-1.81	0.04	-1.76	0.03
2040	0.08	-8.50	-7.17	-3.95	0.06	-2.41	0.03
Average	0.02	-3.65	-4.11	-1.62	0.02	-0.29	0.02

Both these policy scenarios also show a reduced CO₂ intensity compared to BAU. Both these scenarios also show a deterioration in income distribution. Scenario 2 is also showing a lower energy intensity relative to BAU. But, scenario 4 is not energy efficient, because in this scenario, the energy intensity relative to BAU is higher. Overall, scenario 2 can safely be said to be better of the two. It enables the economy to achieve, in comparison to BAU, higher GDP, lower aggregate emissions, lower energy and CO₂ intensities.

None of the recycling options for carbon tax revenue considered here yield the triple dividend of inclusive green growth. But double dividend is realizable, as clearly shown by our policy scenarios. Either, we get a greener economy along with a small rise in the pace of GDP growth, even though with a somewhat diminished inclusiveness (scenario 2), or, we get a greener economy with enhanced inclusiveness, but with a minor slowdown in economic growth (scenario 3).

5.8. Sensitivity analysis

For checking the robustness of the results of our model we conducted a sensitivity analysis with respect to the following four parameters:

- (1) Elasticity of substitution between capital and labour;
- (2) Armington elasticity of substitution between imported and domestic commodities;
- (3) Annual TFP growth rate
- (4) Annual autonomous energy efficiency improvement (AEEI) growth rate.

We increased each of the above parameters by 10 per cent and ran the corresponding BAU and the five scenarios again. That is, we generated six sensitivity scenarios for each of the above parameters. For example, for parameter elasticity of substitution between capital and labor, we ran the following six scenarios: S1-BAU, S1-Policy Scenario:1, S1-Policy Scenario:2, S1-Policy Scenario:3, S1-Policy Scenario:4, S1-Policy Scenario:5.¹³ The percentage deviations of the selected variables in the scenarios with respect to the BAU of our sensitivity simulations turned out to be remarkably similar to those in the original policy scenarios compared to the original BAU, providing confirmation of the robustness of the results of the former. We report here the results of S1-

Policy Scenario:1 and S1-Policy Scenario:2 vis-à-vis S1-BAU in [Tables 9 and 10](#). As can be seen, the results in these two tables are notably similar to those in [Table 3](#) (Policy Scenario 1) and 4 (Policy Scenario 2) respectively.

6. Conclusion and policy implications

The policy scenarios that we have developed in this paper provide clarity about the trade-offs involved in meeting the three favourite objectives of policy makers in emerging economies like India, namely, accelerating growth in GDP, achieving greater inclusiveness through an improved distribution of income and transiting towards a greener economy through carbon emissions abatement, from a market-based policy instrument like carbon tax. The lucidity about the policy choices to be made follows from five key lessons of our scenarios.

The first lesson is that carbon tax, no matter how it is recycled, cannot be a promoter for inclusive green growth. But, a double dividend is possible to achieve in one of the following two ways: (i) more greening along with greater inclusiveness, but for these to be attained, GDP growth would have to be traded off somewhat (scenario 3), or (ii) more greening along with enhanced GDP growth, but then the trade-off would be in terms of a decline in inclusiveness (scenario 2).

The second lesson is that, when carbon tax revenues are recycled specifically towards development of the clean energy sectors there are no perceptible gains in GDP (scenario 4). One reason for this observation could be that capital cost of clean energy generation is higher relative to conventional energy, and hence investments in these sectors are likely to be costly for the economy. Also, energy intensity of GDP rises because clean energy becomes cheaper for the end-user due to lower per unit cost, and, hence, more of it is used. Therefore, investment in clean energy sectors is beneficial only in terms of climate change mitigation, but not in terms of GDP or income distribution. In other words, this scenario suggests that overinvestment in clean energy sectors may be an issue in an economy like India. Overinvestment in clean energy sectors may induce a shift towards energy intensive production, which may not be beneficial for a labor-abundant economy like India.

The third lesson is that fears of major GDP losses from carbon taxes are exaggerated. If there are GDP losses, they are minimal (scenarios 1, 3 and 5). Indeed carbon tax revenues can be recycled into additional investment in all sectors to secure GDP gains (scenario 2).

The fourth lesson is that whenever GDP gains are garnered by recycling carbon tax revenues into additional investment in all sectors, it is not carbon emission abatement which suffers, but inclusiveness gets

¹³ Here, 'S1' means increasing the parameter (1), S2 implies raising parameter (2), and so on, in our sensitivity simulations).

compromised.

The fifth lesson is that it is the policymakers' priorities about growth and inclusiveness which will decide whether the carbon tax revenues are to be recycled exclusively towards additional investment in all sectors (scenario 2), or, whether these revenues are to be shared between additional overall investment and extra cash transfers to households (scenario 3).

Our study could be extended to analyse other market based instruments like tradable emission permits – domestic and international – but data and time constraints did not allow us to conduct such analysis. Therefore, our focus in this study has been only on domestic carbon taxes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Vijay P. Ojha: Formal analysis, Supervision, Validation, Writing - original draft, Methodology. **Sanjib Pohit:** Writing - original draft, Writing - review & editing, Data curation, Methodology. **Joydeep Ghosh:** Resources, Software, Data curation, Visualization, Methodology.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enpol.2020.111708>.

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